APPROXIMATE COUNTING

Use Knapsack as an example, but W can be large (cost $\sim nW$)

<u>IDEA</u>: scale the weights by a factor $K = \frac{W}{n^2}$

$$\tilde{w}_i = \left\lfloor \frac{w_i}{K} \right\rfloor = \left\lfloor \frac{n^2 w_i}{W} \right\rfloor \quad , \quad \tilde{W} = n^2$$

<u>obs</u> this time we scale w_i and change W

① Original instance: v_i, w_i, W

we do not use

② Scaled instance: $v_i, \tilde{w}_i, \tilde{W}$

F = set of all feasible sets in ① $\hookrightarrow \sum_{i \in S} w_i \leq W$

$\tilde{F}$ = set of all feasible sets in ② $\hookrightarrow \sum_{i \in \tilde{S}} \tilde{w}_i \leq \tilde{W}$

obs the cost for solving (2) exactly is $O(n\tilde{W}) = O(n^3)$ time

poly time using DP 1

What about F and (1) ?

obs $F \subseteq \tilde{F}$ (i.e. $\forall S : S \in F \Rightarrow S \in \tilde{F}$)

hp. $S \in F \Rightarrow \sum_{i \in S} w_i \le W$

$K = \frac{W}{n^2}$

$$\sum_{i \in S} \tilde{w}_i \le \frac{1}{K} \sum_{i \in S} w_i \le \frac{1}{K} W \le \frac{n^2}{W} \cdot W = n^2 = \tilde{W} \Rightarrow S \in \tilde{F}$$

$\tilde{w}_i = \lfloor \frac{w_i}{K} \rfloor$

obs $|F| \le |\tilde{F}|$

We now show that $\frac{|\tilde{F}|}{n+1} \leq |F|$, we assume wlog $0 \leq w_0 \leq w_1 \leq \dots \leq w_{n-1} \leq W$

Claim (A) Let layest r : $w_r \leq \frac{W}{n}$

$S \subseteq [n]$

$T \subseteq [r] \Rightarrow T \subseteq F$

$$\sum_{i \in T} w_i \leq \sum_{i \in T} \frac{W}{n} \leq W$$

& $|T| \leq n$

COR (A) $S \in \tilde{F} / F \Rightarrow S \notin [r]$

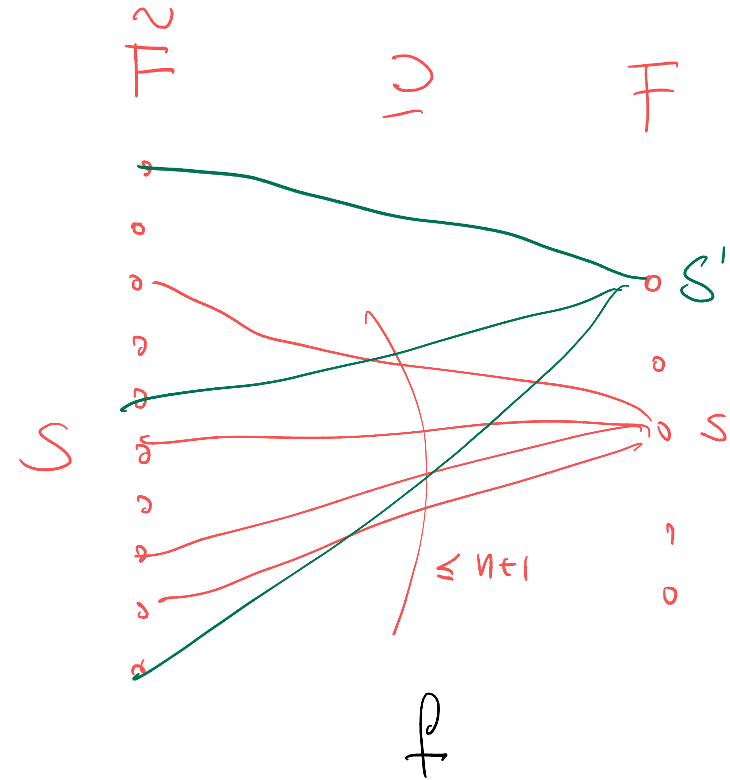
"error"

<u>IDEA</u> to prove $\frac{|\tilde{F}|}{n+1} \le |F|$:

find a mapping $f: \tilde{F} \to F$ such that for each $S' \in F$ there exist at most $n+1$ sets $S \in \tilde{F}$ with $f(S) = S'$

$\Rightarrow \; |\tilde{F}| \le (n+1)\,|F|$

No overcounting



Given $S \in \tilde{F}$, we define

$$f(S) = \begin{cases} S & \text{if } S \in F \\ S \setminus \{e\} & \text{o.w, where } w_e \text{ is the largest weight for } S \end{cases}$$

we remove the element of max weight

obs (B) $\ell > r$ as otherwise $S \in F$ by COR (A)

and thus $w_e > \frac{W}{n}$

claim (C) $f(S) \in F$

$S \in F$: $f(S) = S \in F$, trivial!

$S \notin F \Longrightarrow$ see next

Let $\frac{w_i}{k} = \underbrace{\left\lfloor \frac{w_i}{k} \right\rfloor}_{=\tilde{w}_i} + \delta_i$ $\quad 0 \le \delta_i < 1$ $\quad \delta_i$ is the rounding

$k = \frac{W}{n^2}$

$$w_i = k(\tilde{w}_i + \delta_i) = \frac{W}{n^2}(\tilde{w}_i + \delta_i)$$

$$\sum_{i \in f(S)} w_i \le W \Rightarrow f(S) \in F$$

$f(S) = S \setminus \{e\}$

$$\sum_{i \in f(S)} w_i = \frac{W}{n^2} \sum_{i \in f(S)} (\tilde{w}_i + \delta_i) =$$

$$= \frac{W}{n^2}\left[\sum_{i \in f(S)} \tilde{w}_i + \sum_{i \in f(S)} \delta_i\right] =$$

$$= \frac{W}{n^2}\left[\underbrace{\left(\sum_{i \in S} \tilde{w}_i\right)}_{\le \tilde{W} = n^2} - \tilde{w}_e + \underbrace{\left(\sum_{i \in S} \delta_i\right)}_{< |S| \le n} - \delta_e\right]$$

$S \in \tilde{F}$

$w_e \ge \frac{W}{n}$

$$\le \frac{W}{n^2}\left[n^2 - \tilde{w}_e + n - \delta_e\right] = W + \frac{W}{n} - w_e \le W$$

Q. How many distinct sets $S \in \tilde{F}$ maps to $S' \in F$? (i.e. $f(S) = S'$)

For these $S \in \tilde{F} \setminus F$, $f(S) = S \setminus \{e\} \in F$

- 1 choice if $S \in F$
- $\leq n$ choices for e to get $S \setminus \{e\} \in F$

≤ n choices for this element

$\Rightarrow \leq (n+1)$

Summing up... we can estimate $|F|$ using $|\tilde{F}|$ (and DP 1)
in $O(n^3)$ time: $\frac{|\tilde{F}|}{n+1} \leq |F| \leq |\tilde{F}|$

QED

Can we do any better? YES : SAMPLING

FPRAS = FPTAS + randomization (ε, δ)

approx. error

ε-approximation with error probability $\leq \delta$

polynomial time in $n, \frac{1}{\varepsilon}, \log\frac{1}{\delta}$ (e.g. CM sketch)

IDEA: Estimate $p = \frac{|F| \leftarrow \text{unknown}}{|\tilde{F}|}$ using SAMPLING

known $\rightarrow |\tilde{F}|$

(we know that $\frac{1}{n+1} \leq p \leq 1$)

a) Repeat for $t = 1, 2, \ldots, N$: $\leftarrow$ to be chosen

1) random sample a feasible set $S \in \tilde{F}$ (we saw this last class)

2) $X_t = \begin{cases} 1 & \text{if } S \in F \\ 0 & \text{o.w.} \end{cases}$ $\quad \Pr(X_t = 1) = p$

b) Let $X = \sum_{t=1}^{N} X_t$ $\quad (\mu = E[X] = N \cdot p)$

c) Return $z = \frac{|\tilde{F}| \cdot X}{N}$ $\leftarrow$ we know how to compute in $O(n^3)$ time

Running time: $O(\underbrace{n^3}_{|\tilde{F}| \text{ DP 1}} + N \underbrace{n}_{\text{random sample}}) = O(n (N + n^2))$

It only remains to see that Z is an unbiased estimator:

$$E[Z] = \frac{|\tilde{F}|\, E[X]}{N} = |\tilde{F}| \cdot p = |F|$$

$$Z = \frac{|\tilde{F}| \cdot X}{N} \qquad \mu = E[X] = Np \qquad p = \frac{|F|}{|\tilde{F}|}$$

We use Chernoff bound (we saw this in some classes on sketches)

obtained by dividing by $|\tilde{F}|/N$

$$Pr\left[\left|\tilde{Z}-E[\tilde{z}]\right|>\varepsilon E[\tilde{Z}]\right]=Pr\left[|X-\mu|>\varepsilon\mu\right]\le 2e^{-\frac{\varepsilon^2\mu^{\cancel{2}}}{3\cancel{\mu}}}=2e^{-\frac{\varepsilon^2Np}{3}}\le\delta$$

$E[\tilde{z}] = |F|$, $E[\tilde{Z}] = |F|$

$\tilde{z}=\boxed{\frac{|\tilde{F}|}{N}}\cdot X$

$E[X]=\mu=Np$

$$\frac{2}{\delta}\le e^{\frac{\varepsilon^2Np}{3}}\overset{\ln()}{\Longrightarrow}\ln\left(\frac{2}{\delta}\right)\le\frac{\varepsilon^2Np}{3}\Leftrightarrow N\ge\frac{3\ln 2/\delta}{\varepsilon^2p}$$

$$\frac{1}{n+1}\le p\le 1\quad\Rightarrow\quad N=\frac{3(n+1)\ln 2/\delta}{\varepsilon^2}=O\left(\frac{n\ln 1/\delta}{\varepsilon^2}\right)$$

$\Rightarrow$ TOTAL RUNNING TIME $= O\left(n^3+n^2\cdot\log\delta^{-1}\,\varepsilon^{-2}\right)$